Kartik Chandra

Bangalore — Mumbai | +91 70217 68224 | kartiksatishchandra@gmail.com | github.com/kartik-sc | Linkedin

Objective

CS undergrad at RVCE with interests in AI/ML, backend engineering, robotics and systems programming. Interested in and passionate about building AI-driven and system-oriented applications to solve real-world problems.

Education

RV College of Engineering — Bengaluru, India
B.E. in Computer Science and Engineering — *Sept 2024 – Aug 2028*

- **CGPA**: 8.38 / 10.0
- **Relevant Coursework**: C, Python, Data Structures and Algorithms, Operating Systems, Computer Networks + CCNA-1 certification course, Linear Algebra, Statistics, Internet of Things and Embedded Computing

Technical Skills

Languages: C, C++, Java, Python, SQL, TypeScript
Frameworks: FastAPI, Next.js 15, React, LangGraph, SQLAlchemy, Streamlit
Tools & Infrastructure: Git, Linux, PostgreSQL, Redis, Docker, D3.js, NumPy, Pandas, Matplotlib
Domains: Backend Engineering, REST APIs, Full-Stack Development, Multi-Agent AI Systems, Data Structures & Algorithms · System Design
LeetCode Rating: 1703

Projects

AI Research Agent
Python · FastAPI · LangGraph · Next.js 15 · PostgreSQL · Redis · TypeScript · D3.js

- Reduced multi-source research synthesis from several hours to **under 3 minutes** by orchestrating a **5-agent LangGraph pipeline** (Planner → Researcher → Critic → Writer → Memory) with shared Redis context and real-time SSE output streaming to a Next.js 15 dashboard.
- API spend bounded by **caching research responses in Redis with TTL expiry** and slowapi rate limiting on FastAPI, eliminating redundant Gemini calls for repeated queries.

Multi-Threaded Web Crawler
C++17, Multithreading, libcurl, CMake, Graph Algorithms, PageRank

- Developed a high-performance C++17 **web crawler using multithreading**, mutexes, condition variables, thread-safe queues, and libcurl.
- Engineered duplicate detection, scalable crawl scheduling, hyperlink graph construction, and edge-based PageRank using Power Iteration.
- Result: **3.8× speedup**, **95% parallel efficiency**, and **6.7× throughput scaling** on 8 threads.

DupeScan
Python · FastAPI · React · TypeScript · xxHash · SHA-256

- **7,900× faster than brute force**: three-stage pipeline (size grouping → xxHash fingerprint → SHA-256 confirmation) on a 10,000-file corpus, **short-circuiting 99.7% of redundant comparisons**
- FastAPI backend + React/TypeScript frontend: directory path input, live scan progress streaming, duplicate cluster view with per-file deletion controls
- **Fully test-covered** via pytest: all three stages validated against empty files, zero-byte duplicates, and hash collision edge cases.

Predictive Maintenance System
Python, Streamlit, Machine Learning

- Developed a machine learning system for failure prediction using One-Class SVM, in cutting, milling, lathe machines using sensor data.
- End-to-end ML pipeline in scikit-learn: data loading → preprocessing → feature engineering → training → evaluation, with a Streamlit dashboard for real-time sensor input and per-class failure probability output
- Achieved **91% prediction accuracy** on machine failure classification.

Experience

Technical Team Member – Astra Robotics RVCE, Bengaluru
Robotics Projects and Research *2024 – Present*

- Working on bionic arm movement using reinforcement learning, McKibben muscles, and pneumatic actuation.

Member – Google Developer Groups (GDG) RVCE, Bengaluru
Open Source & Competitive Programming *2025 – Present*

- Developing an open-source competitive programming platform using Electron.js, Redis, WebAssembly, Node.js, and MongoDB.